

Impedance Matching Lab Report

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SUMMARY

Impedance matching is critical for understanding signal propagation in cables and taking accurate measurements in experimental set ups. Here, we empirically investigate the effect of terminators of different resistances on a reflected signal to understand the effect of incorrectly-terminated cables. We find that at 50Ω the reflected signal is appropriately damped by the matched impedance, which agrees with the expected value from our cable's manufacturer. We also qualitatively assess unwanted effects from combining cables of different impedance.

1 INTRODUCTION

Modern physics experiments rely on accurate signal transmission and processing to measure relevant quantities. Understanding the nature of signal propagation and behavior is therefore critical to making scientific measurements. One unwanted effect arises when a circuit is improperly terminated, resulting in a signal reflection. Matching the cable's impedance, or characteristic ratio of voltage to current along a cable [1], is important for ensuring accurate measurements and the most power transmitted in the signal.

Because impedance is intrinsic to the cable, measuring the amplitude of reflection at different resistances can be used to empirically determine the impedance of the cable. Here, we investigate the characteristic impedance of the RG-58 cable to understand the effect of reflected signals on signal propagation and determine the characteristic impedance of the cable.

2 THEORY

Signals propagate as electromagnetic waves through cables, and therefore obey Maxwell's laws. Recall that the intrinsic impedance Z of a cable relates the voltage to the current along the cable. From Kirchoff, we know that $V = IR$ which relates the resistance to the quantity of interest. With $Z = V_r/I_r$ for the reflected voltage and current and $R = V_L/I_L$ for the voltage and current about the load, one can derive the equation for the impedance using continuous boundary conditions.

Consider a signal of amplitude A_o traveling down a cable with characteristic impedance Z . If the cable is not terminated at all, then the full amplitude A_o is reflected opposite to the original signal with the delay time corresponding to the propagation speed inside the cable, similar to standing waves in an open-ended tube. Similarly if the cable is terminated with a resistor of resistance R but the impedance not properly matched, the amplitude A_r of the reflected waveform is given by,

$$A_r = \frac{R - Z}{R + Z} A_o \quad (1)$$

This equation can be derived from Maxwell's Equations. If the resistance is small but greater than 0, the reflected wave form will point the same direction as the true signal. However, if $R > Z$ then the waveform of the reflection will peak in the opposite direction to the true signal, i.e. upside down. We can measure the polarity of the reflection with the parameter Γ where

$$\Gamma = \frac{A_r}{A_o} = \frac{R - Z}{R + Z} \quad (2)$$

If $\Gamma > 1$ then the reflection peaks in the same direction as the signal, and if $\Gamma < 1$, the reflection is inverted. The ideal case occurs when $\Gamma = 0$ meaning that all energy from the true signal is delivered to the resistor. In this case, there is no reflection to interfere with the signal and the maximum amount of power is extracted from the system.

Equation 1 allows us to empirically determine Z by measuring the voltage of the reflected signal for different resistances and fitting for parameters A_o and Z .

3 EXPERIMENTAL METHODS

Following the experimental set up shown in Figure 1, we attach the Phillips Pulser to the oscilloscope using a short RG-58 cable, two BNC tee's and a 50Ω terminator at the port on the tee connected to the oscilloscope. We connect a 20ft RG-58 cable which is initially left open at one end.

We use a voltmeter to measure the resistances of the resistors used in this experiment which are detailed in Appendix A. We then use the oscilloscope to measure the amplitude of the reflected signal from peak to peak. We make five measurements and calculate the average and standard deviation. We attach each resistor to the end of the 20ft RG-58 cable and repeat the measurement. These measurements are also detailed in Appendix A

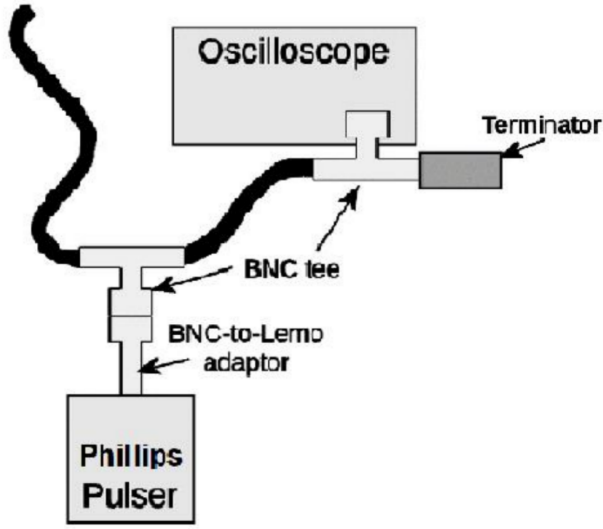


Figure 1. Experimental set up used to measure the characteristic impedance of RG-58 cables. The unterminated cable in the top left corner is terminated with resistors of varying resistances, and we use the oscilloscope to measure the amplitude of the reflected waveform determined by the Phillips Pulser.

4 DATA ANALYSIS AND RESULTS

We use CURVE FIT from SciPy’s Optimize package to fit the reflection amplitude as a function of resistance. We incorporate errors in both variables and find the best fit parameters to Equation 1, which is shown by the blue

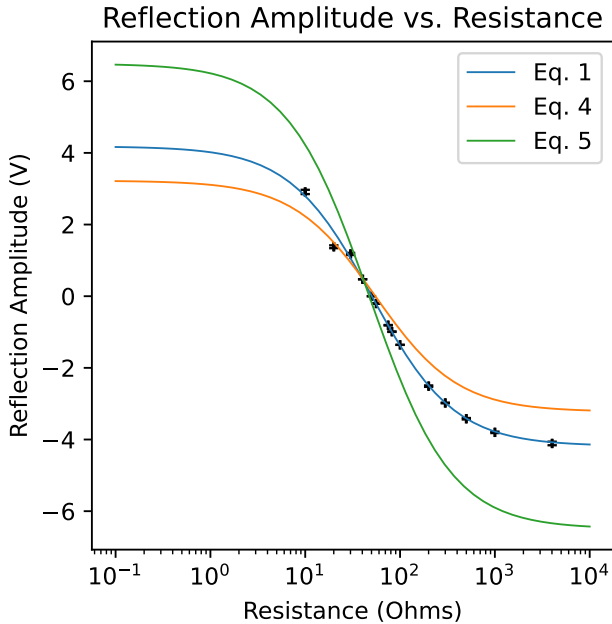


Figure 2. The amplitude of the reflected signal as a function of resistance used to terminate the RG-58 cable. We fit for the parameters using three different equations, and find the lowest χ^2 fit from Equation 1 as expected.

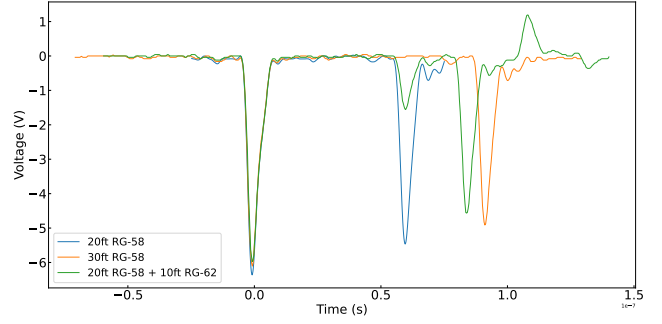


Figure 3. Waveforms recorded by the oscilloscope for open (unterminated) cables. The figure shows 20ft of RG-58 (blue), 30ft of RG-58 (orange) and 20ft of RG-58 combined with 10ft of RG-62 (green).

line in Figure 2. The green and orange lines are determined by the equations $A = (R^2 - Z^2)/(R^2 + Z^2)A_o$ and $A = (\sqrt{R} - \sqrt{Z})/(\sqrt{R} + \sqrt{Z})A_o$, respectively. It is visually apparent that Equation 1 is the correct relationship between resistance and reflected voltage amplitude, as the other equations do not accurately capture the behavior at the tails of the distribution. The blue line has a reduced χ^2 fit of 9.78, indicating that our errors are underestimated. However, this is significantly better than the green and orange lines which have reduced χ^2 fits of 279.0 and 144.6, respectively.

The best fit parameters for the blue curve are $Z = 50.4 \pm 0.6 \Omega$ and $A_o = -4.18 \pm 0.04$ V. The impedance agrees with the expected value of 50Ω within its 1σ uncertainty. We measured a value $A_o = -4.11$ V for the amplitude of the signal before reflection, so we are within 2σ agreement of the expectation.

5 EXPLORING MIXED CABLES

In Figure 3, we show the effect on the reflected signal when cables of different impedance are used. The blue line is the waveform for 20ft of RG-58 left unterminated, as we examined in Section 4, and the orange line shows the same for 30ft. The wave shows the main pulse on the left side, which all three lines show, then a second smaller peak, whose amplitude is given by Equation 1, separated by the signal propagation time. However, the green curve, which shows the reflection for 20ft of RG-58 cable and 10ft of RG-62 cable, has a more complex shape. The main pulse and its reflection are still clearly visible, but a second, smaller pulse and reflection with opposite polarity are also visible. It is also interesting to note that the peak from the mixed wires occurs between the 20ft and 30ft of RG-58. These observations suggest that the RG-62 has a faster propagation speed than RG-58, and that using cables with different impedance produces a second polarized reflection.

The blue and orange curves have peaks caused by the effect we examined in Section 4, where the main signal is simple reflected from the open end of the cable like a standing wave. However, we see interesting behavior in the green curve caused by the differing impedance. The main peak is the true signal pulse, and the next, smaller peak is the reflection at the boundary between the RG-58 and RG-62

Peak	Pred Arr Time (ns)	Act Arr Time (ns)
Blue 1	0.0	-0.894
Blue 2	60.61	59.57
Orange 1	0.0	-0.800
Orange 2	90.91	91.07
Green 1	0.0	-0.850
Green 2	60.61	59.78
Green 3	84.70	83.84
Green 4	108.80	107.90

Table 1. Predicted and measured arrival times for each peak shown in Figure 3. We describe the peak by the color of the curve in the Figure and number the peaks from left to right. The cause of each peak is described in Section 5.

cable. This agrees with the propagation time we see in the blue curve (the same 20ft of RG-58 cable) but has a smaller amplitude because most of the signal is transmitted through to the RG-62 cable. The next peak is the reflection from the end of all 30ft of cable. The fourth peak appears to be a reflection of the second peak with opposite polarity. I hypothesize that this signal travels to the end of the 30ft of cable, is reflected at the boundary with the RG-58 cable, travels again to the open boundary, and then is then registered by the oscilloscope. Since we know that the second and third peaks are separated by 20ft of cable propagation time, and the third and fourth appear to be spaced the same, this accounts for 20ft of propagation.

The predicted and actual arrival times are listed in Table 1. The predicted arrival time is calculated by dividing the length of cable by the speed of signal propagation in that cable given by the manufacturers. For the RG-58 cable this is $0.66c$ [2] and $0.83c$ [3]. Uncertainties arise from the precision of the recorded oscilloscope data, the precision of the length of the cable, and the measurement of the propagation speed from the manufacturer.

6 CONCLUSIONS

We used resistors of varying resistances to empirically determine the characteristic impedance of an RG-58 cable. We recorded the resistance and reflected signal amplitude to test the relationship given by Equation 1. We find good agreement with the expected values in our measurements. We have examined the importance of understanding cable impedance and appropriately terminating cables for signal propagation and measurement.

7 REFERENCES

- [1] Class 4 Notes, Chris Hill, Carmen/Canvas, 2025.
- [2] Introductory Laboratory III: Impedance Matching, Carmen/Canvas, 2024.
- [3] 48" BNC Coaxial Cable, BNC Male to BNC Male, Thorlabs, 2021.
- [4] 93 Ohm Flexible RG62 Coax Cable Single Shielded with Black PVC Jacket, Pasternak.

Expected R (Ω)	Measured R (Ω)	Reflection Amplitude (V)
0	0.0 ± 0.1	
10	10.0 ± 0.1	2.91 ± 0.06
20	20.0 ± 0.2	1.38 ± 0.04
30	30.1 ± 0.3	1.18 ± 0.03
40.2	40.3 ± 0.4	0.473 ± 0.003
49.9	49.9 ± 0.5	$0 \pm 0^\dagger$
56	56.0 ± 0.5	-0.204 ± 0.008
75	75.0 ± 0.7	-0.810 ± 0.007
82	82.1 ± 0.7	-0.990 ± 0.007
100	100.1 ± 0.9	-1.358 ± 0.001
200	200.2 ± 1.8	-2.51 ± 0.02
300	300.1 ± 2.7	2.98 ± 0.01
499	500.2 ± 4.5	-3.42 ± 0.02
1000	1001 ± 9	-3.80 ± 0.02
4020	4021 ± 36	-4.11 ± 0.05

Table A1. Expected resistances, measured resistances, and the amplitude of the reflected signal using the resistor to terminate the RG-58 cable. Measurement uncertainties for the resistor combine estimates from manufacturing and measurement errors. The uncertainties for the signal amplitude are the standard deviation of five independent measurements. † We were unable to measure an amplitude or uncertainty because the waveform was not distinguishable from the noise in the oscilloscope.

APPENDIX A: EXPERIMENTAL MEASUREMENTS

The resistance measurements for the resistors used in this experiment are given in Table A1. The measurement uncertainties include the systematic uncertainty from the manufacturer with the measurement uncertainty from our procedure. We also detail the measured amplitude of the reflected signal with its associated uncertainty. Note that a negative measurement indicates that the waveform is inverted from the true signal, and that positive means the maximum amplitude points in the same direction as the signal.